CSC 461. Individual Project Proposal

Predicting Maintenance Needs from Cessna 172 Flight Data: Is the Model Seeing Wear or Just Context?

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Purpose, Motivation, Impact

General aviation (small private, training and recreational aircraft) is one of the most dangerous parts of civil aviation, according to the NTSB (cited in the dataset paper). In 2023 the AOPA Air Safety Institute (AOPA, 2025) counted 1,097 U.S. general aviation accidents, 186 of them fatal, which is a fatal accident rate of 0.65 per 100,000 flight hours (28.4 million hours flown). About 70% of the 929 non-commercial fixed-wing accidents were pilot-related, but mechanical-related accidents still numbered 187, and fatal ones rose from 21 in 2022 to 33 in 2023. Unexpected repairs also ground aircraft and cost money. The Cessna 172 is the most produced aircraft (over 43,000 built, according to the dataset paper), so a method that works for it can help many operators.

This project uses a public dataset of real Cessna 172 flights and maintenance logs (NGAFID) to predict whether a flight happened shortly before an unplanned repair. I do not claim that this prevents accidents; the goal is to find out how reliable such predictions really are. Published models reach about 76% accuracy, but the dataset authors (Yang and Desell, 2022) warn that pilot actions, weather and flight plans explain much of the variation in the data, not only the condition of the aircraft.

My questions:

- **RQ1, wear or context?** Does a model use signs of engine problems, or does it use information about the day and the flight (weather, fuel, flying style)?

- **RQ2, honest testing.** Do results drop when flights from the same repair event cannot appear in both training and test data, or when a whole type of problem is hidden from training?

- **RQ3, which sensors?** Which sensor groups does the model need, and does that make sense physically?

- **RQ4, pilot or aircraft?** How much of the variation in the sensors is explained by how the flight was flown (phase of flight, speed, altitude, weather), and how much is left over? Is that leftover part what allows a model to tell flights before a repair from flights after it?

- **Cost.** The data has no cost information, so I will not claim savings. I will show for which ratios between the cost of a missed problem and the cost of a false alarm a model is worth using.

Dataset

NGAFID Aviation Maintenance Dataset (Yang and Desell, Rochester Institute of Technology, 2022; public on Zenodo and Kaggle). The flight data come from the National General Aviation Flight Information Database (NGAFID). A single flight school recorded them automatically after each flight over several years, and the maintenance logs come from the MaintNet project (Akhbardeh et al., 2020). It has 28,935 real (not simulated) flights, 31,177 hours, all on Cessna 172 aircraft, linked to 2,111 unplanned maintenance events of 36 types. Each flight has 23 sensors recorded every second (over 100 million rows, 4.3 GB). The authors also give a balanced benchmark subset: 5,602 flights shortly before a repair and 5,844 shortly after, covering 19 issue types.

Can it answer my questions? Yes for one aircraft model and one flight school; it cannot show whether results transfer to other aircraft. Also, every flight comes from the days just before or after a repair, so the model never sees a "healthy" aircraft: it only learns to tell flights before a repair from flights after it. A

difference between the two can come from the repair itself and not only from wear, which is what RQ1 and RQ4 try to separate. Aircraft serial numbers and flight dates were removed for privacy, so I cannot split by aircraft or by time. I use only this dataset and I will not collect data (I have no aircraft or simulator, and flying faulty aircraft would be unsafe). There is no second public dataset like this one that I know of.

Methods

Input Data

The data are multivariate time series: each flight is a table with one row per second and 23 sensor columns, so flights have different lengths. The sensors are electrical (volt1, volt2, amp1, amp2), fuel (left and right quantity, fuel flow), engine (oil temperature and pressure, RPM, 4 cylinder head and 4 exhaust gas temperatures) and context (outside air temperature, airspeed, vertical speed, acceleration, altitude). About 1% of the values are missing. As in the benchmark, I keep the last 4,096 seconds (about 68 minutes) of each flight. For the course models (logistic regression, random forest, k-NN) I turn each flight into summary numbers per sensor (mean, standard deviation, minimum, quartiles, maximum). For the neural network I use the raw sequences. Missing values and scaling use training data only; the benchmark scaled with the minimum and maximum of all data, which I will avoid.

Splitting. First, the 5 folds given by the authors (to compare with their results). Then stratified 5-fold cross-validation with extra checks: (a) grouping by maintenance event, if the data contain an event identifier (to be checked first, when I download the data), and (b) leaving whole issue types out of training. Hyperparameters are tuned only on a validation part of the training data, and the part held out is the test set.

Target Variable(s)

A class (classification). Main task: binary detection, positive = flight within the two days before an unplanned repair, negative = flight after it. Secondary tasks: classify the 19 issue types, or the 5 groups the authors define (engine, baffle, oil, cylinder, other), mostly to study confusions.

Model

I compare a few models on the same splits, from simple to complex:

- **Logistic regression** and **random forest** on summary features (taught in class, easy to interpret, and the forest gives variable importance).

- **Linear regression**, used in RQ4 to remove the part of the engine readings that the context sensors explain.

- **InceptionTime** (Ismail Fawaz et al., 2020), a 1D convolutional network made for time series, using the dataset authors' public code; it reached 75.5% in their paper.

- Optional: **k-NN**, MiniRocket (Dempster et al., 2021) and ConvMHSA (Yang et al., 2021), the authors' transformer-style model (convolutions followed by 4 self-attention layers). It is only half a point better than InceptionTime in their paper (76.0% vs. 75.5%), so it is not essential for my questions.

First I reproduce the published 5-fold results as a check. The authors trained on a TPU, so the neural network is the expensive part: I will use fewer epochs on free Colab GPUs and run the heavy model only where needed.

Metrics and Results

Metrics. ROC curves and AUROC, confusion matrices, precision, recall and F1, and accuracy only to compare with the paper (the binary subset is balanced, so accuracy is fair there). For the 19-class task the classes are very imbalanced, so I will use per-class recall and macro-F1 instead. I report mean +/- SD over folds and bootstrap confidence intervals.

What I will do for each question:

- RQ1, wear or context? I train the same model three times: with all sensors, with engine sensors only, and with context sensors only (groups defined in Input Data). If the context-only model gets close to the full one, the model is mostly reading the situation, not the engine. I also plot recall against the number of flights before the repair, to see whether earlier flights are detected or only the last ones. Figures: AUROC by sensor set; recall vs. flights before repair.

- RQ2, honest testing. I run the same models under three splits: the authors' folds, folds grouped by repair event (if an event identifier exists), and leave-issue-types-out (test on issue types never seen in training). Figure: AUROC per split with confidence intervals. A drop means the first results were optimistic.

- RQ3, which sensors? I remove or shuffle one sensor group at a time (permutation importance for the random forest, group ablation for the other models) and measure the drop in AUROC. Physical check: a model for oil problems should rely on oil sensors, and one for cylinder problems on cylinder temperatures; if it relies on altitude or air temperature, it is reading context. Figure: AUROC drop per sensor group and issue group.

- RQ4, pilot or aircraft? I use linear regression, fitted on training flights only, to explain the engine temperatures and oil readings with the context sensors plus RPM and fuel flow, which the pilot controls. The part left over is "not explained by the recorded context". I compare models trained on the raw data, on the context sensors only (the same model as in RQ1) and on the residuals only. Figures: variance explained per sensor; AUROC by type of input. Limits: there is no pilot identifier, some pilot actions (such as the mixture setting) are not recorded, and the leftover part is not proof of a mechanical cause.

- Cost. I change the decision threshold of the best model and count missed problems and false alarms at each threshold. For each cost ratio R (cost of a missed problem / cost of a false alarm) I plot the expected cost and mark where the model beats "never alert" and "always alert".

External data: I will not test on an external dataset, because I know of no other public dataset with real flights and maintenance logs of this kind (simulated engine datasets use different engines, sensors and tasks). Leave-issue-types-out is my closest check of generalization, and I will not claim the results transfer to other aircraft or flight schools.

Priority: RQ1 and RQ2 are the core of the project; RQ3 reuses the experiments of RQ1; RQ4 and Cost are extensions if time allows.

Overlap statement

For now this project is not related to my thesis and does not overlap with a project for another class. I chose it because of my personal interest in applying computer science and engineering to aviation.

Reflection

References

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